Logistic Regression and Classification

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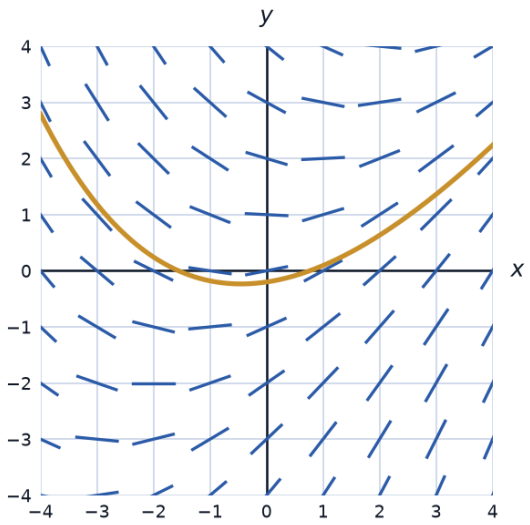
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 60$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

2 [Foundational] • Target: The Odds Ratio and Logit Formulation

In evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.



3 [Foundational] • Target: Binary Cross-Entropy Loss

In evaluating a binary classifier, the confusion matrix shows $TP = 45$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

4 [Foundational] • Target: The Confusion Matrix and Core Metrics

In evaluating a binary classifier, the confusion matrix shows $TP = 55$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

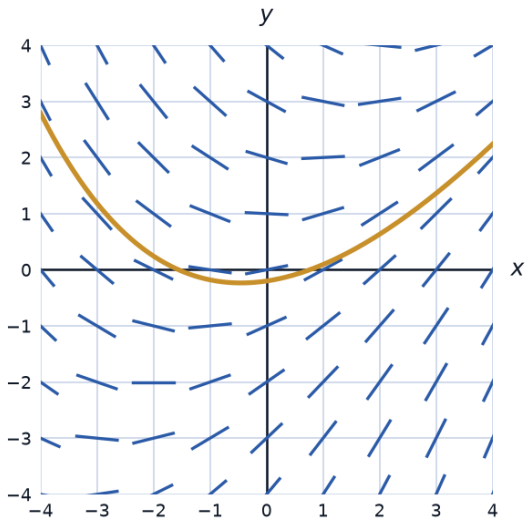
5 [Foundational] • Target: The F_1-Score, Thresholding and ROC-AUC

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.30$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

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6 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.40$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.



7 [Foundational] • Target: The Odds Ratio and Logit Formulation

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.50$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Binary Cross-Entropy Loss

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.60$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

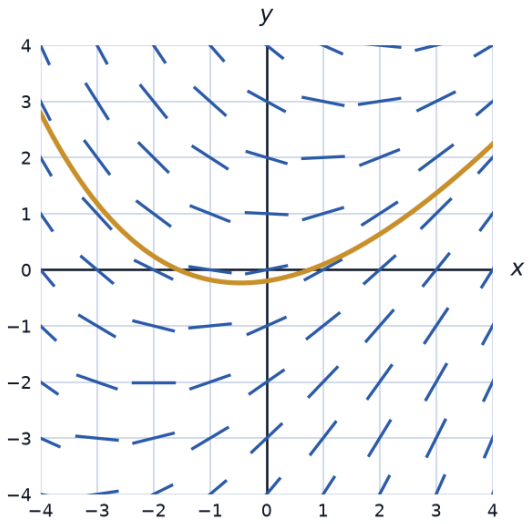
9 [Intermediate] • Target: The Confusion Matrix and Core Metrics

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.70$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

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10 **[Intermediate]** • **Target: The F_1-Score, Thresholding and ROC-AUC**

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.80$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.



11 **[Intermediate]** • **Target: Logistic Regression & Classification Practice Exercises**

Consider a binary classification problem where the decision boundary is defined by the logistic function $p(x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2)}}$, which represents the probability of a data point $\mathbf{x} = (x_1, x_2)$ belonging to class 1. The estimated model parameters are $\beta_0 = -3$, $\beta_1 = 1$, and $\beta_2 = -2$. We are interested in the geometric shape of the classification boundary. The classification boundary occurs where $p(x) = 0.5$. Find the equation of this boundary in the x_1x_2-plane.

12 **[Intermediate]** • **Target: The Odds Ratio and Logit Formulation**

A binary classification model uses a logistic sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$. The decision boundary is defined by $\sigma(z) = 0.5$, where $z = w_1x_1 + w_0$. For a dataset with two features x_1 and x_2, the model's linear component is $z = 2x_1 - x_2 + b$. If the classification boundary passes through the point $(x_1, x_2) = (3, 1)$, what is the value of the bias b? Show your complete algebraic work.

13 **[Intermediate]** • **Target: Binary Cross-Entropy Loss**

A spam classification model identifies $TP = 120$ genuine spam emails and $FP = 40$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

14 **[Intermediate]** • **Target: The Confusion Matrix and Core Metrics**

A spam classification model identifies $TP = 150$ genuine spam emails and $FP = 50$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

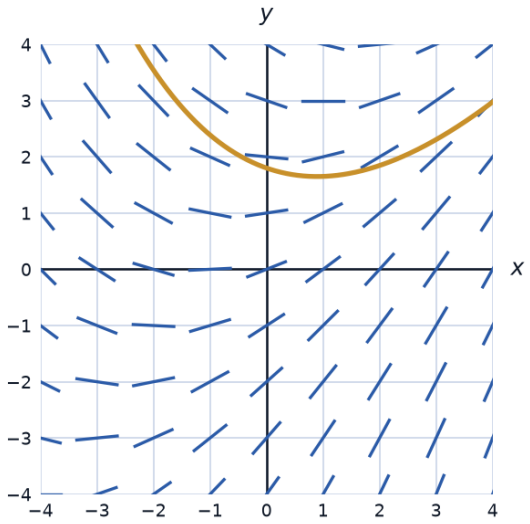
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15 [Challenging] • Target: The F_1-Score, Thresholding and ROC-AUC • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A spam classification model identifies $TP = 180$ genuine spam emails and $FP = 60$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

16 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

A spam classification model identifies $TP = 90$ genuine spam emails and $FP = 30$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.



17 [Challenging] • Target: The Odds Ratio and Logit Formulation • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

A spam classification model identifies $TP = 30$ genuine spam emails and $FP = 10$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

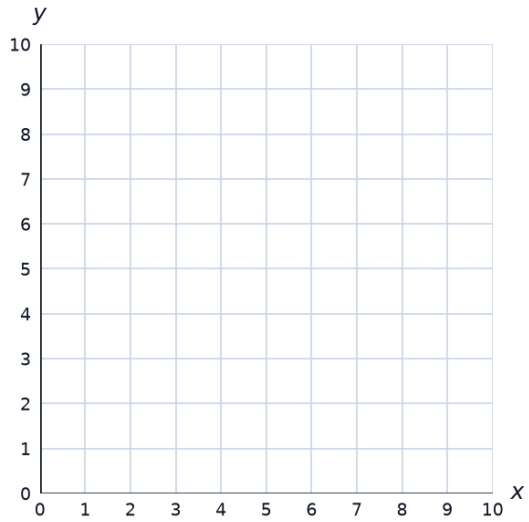
18 [Challenging] • Target: Binary Cross-Entropy Loss • Cross-Lesson: 3.01 Gradient Descent and Optimization

A spam classification model identifies $TP = 60$ genuine spam emails and $FP = 20$ legitimate emails falsely flagged as spam. What is the precision of this model? Show your complete algebraic work.

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19 [Challenging] • Target: The Confusion Matrix and Core Metrics • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

The scatterplot below displays the relationship between the number of hours students spent studying for a test and their scores on that test. The equation of the least-squares regression line is approximately $\hat{y} = 55.2 + 4.8x$, where x is the number of hours spent studying and $\hat{y}$ is the predicted test score.



20 [Challenging] • Target: The F_1-Score, Thresholding and ROC-AUC • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A social scientist is investigating the relationship between the number of hours spent studying per week (X) and the final exam score (Y) for a group of $n = 100$ students. The least-squares regression line is given by $\hat{y} = 55.2 + 8.5x$. The standard error of the slope is $SE_b = 1.2$. The researcher wants to construct a 95% confidence interval for the true slope of the regression line. What is the value of the t-score used for this confidence interval? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 45$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

a. 81.0%

b. 81.8%

c. 52.4%

d. 13

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22 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows \(TP = 50 \), \(TN = 40 \), \(FP = 10 \), and \(FN = 10 \). Compute the overall classification accuracy.

a. \(81.8\% \)

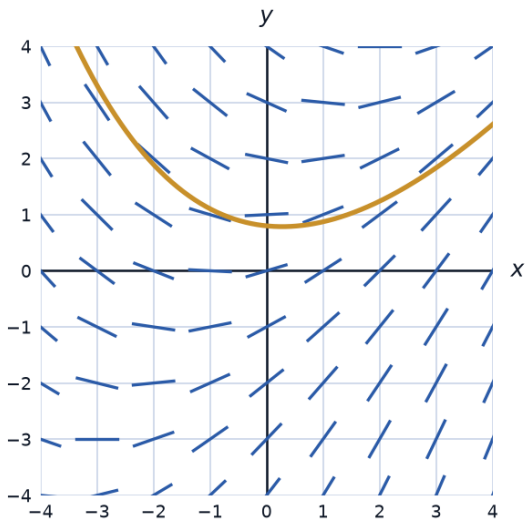
b. \(83.3\% \)

c. \(54.5\% \)

d. \(13 \)

23 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows \(TP = 55 \), \(TN = 40 \), \(FP = 10 \), and \(FN = 10 \). Compute the overall classification accuracy.



a. \(82.6\% \)

b. \(84.6\% \)

c. \(56.5\% \)

d. \(13 \)

24 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows \(TP = 60 \), \(TN = 40 \), \(FP = 10 \), and \(FN = 10 \). Compute the overall classification accuracy.

a. \(83.3\% \)

b. \(85.7\% \)

c. \(58.3\% \)

d. \(13 \)

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25 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 65$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

a. 84.0%

b. 86.7%

c. 60.0%

d. 13

26 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

a. 84.6%

b. 87.5%

c. 61.5%

d. 13

27 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

A spam classification model identifies $TP = 30$ genuine spam emails and $FP = 10$ legitimate emails falsely flagged as spam. What is the precision of this model?

a. 75.0%

b. 60.0%

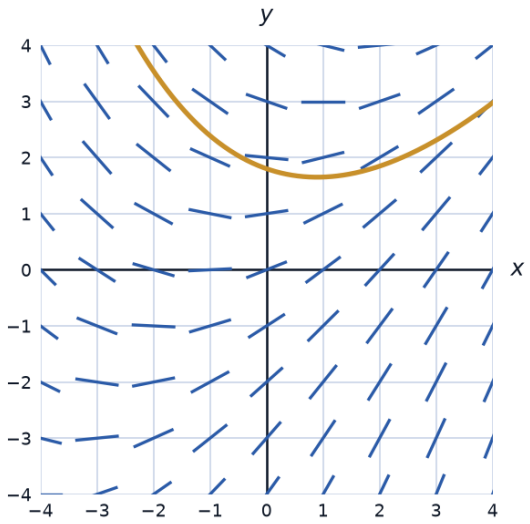
c. 25.0%

d. 85.7%

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28 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

A spam classification model identifies \(TP = 60 \) genuine spam emails and \(FP = 20 \) legitimate emails falsely flagged as spam. What is the precision of this model?



a. \(75.0\% \)

b. \(60.0\% \)

c. \(25.0\% \)

d. \(85.7\% \)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Odds Ratio and Logit Formulation

If a classifier predicts probability $p = 0.80$ for disease occurrence, what are the odds of having the disease?

a. 4.0

b. 0.25

c. 0.80

d. 5.0

30 [Intermediate] • Target: The Odds Ratio and Logit Formulation

Compute the log-odds (logit) value when probability $p = 0.50$.

a. 0.0

b. 1.0

c. 0.5

d. -1.0

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31 [Intermediate] • Target: The Odds Ratio and Logit Formulation

If a logistic model has equation $\ln\left(\frac{p}{1-p}\right) = 2.0$, find the predicted probability p.

a. $\frac{e^2}{1+e^2} \approx 0.88$

b. $\frac{1}{2} = 0.50$

c. $\frac{2}{3} \approx 0.67$

d. 0.95

32 [Intermediate] • Target: Binary Cross-Entropy Loss

Calculate the binary cross-entropy loss $L = -[y \ln \hat{y} + (1-y)\ln(1-\hat{y})]$ for a sample where true label $y = 1$ and predicted probability $\hat{y} = 0.90$.

a. $-\ln(0.90) \approx 0.105$

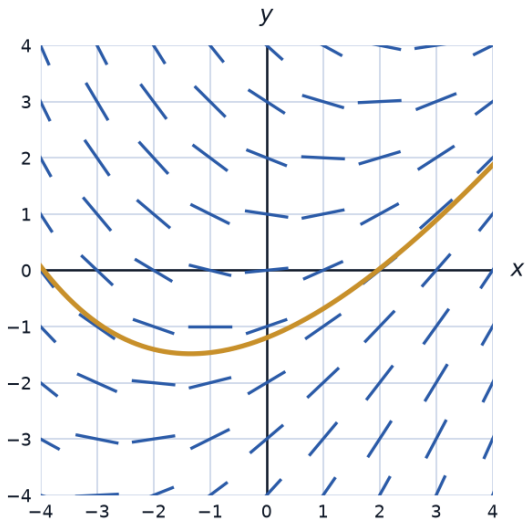
b. $-\ln(0.10) \approx 2.30$

c. 0.90

d. 0.0

33 [Intermediate] • Target: Binary Cross-Entropy Loss

What is the derivative of the binary cross-entropy loss with respect to the pre-activation logit z, where $\hat{y} = \sigma(z)$?



a. $\hat{y} - y$

b. $\hat{y}(1-\hat{y})$

c. $y - \hat{y}$

d. $\frac{\hat{y}-y}{\hat{y}(1-\hat{y})}$

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34 [Intermediate] • Target: Binary Cross-Entropy Loss

What happens to the cross-entropy loss as $\hat{y} \to 0$ when true label is $y = 1$?

a. It approaches $+\infty$, heavily penalizing confident wrong predictions

b. It approaches zero

c. It oscillates around 1

d. It equals the pre-activation logit

35 [Intermediate] • Target: The Confusion Matrix and Core Metrics

A diagnostic model reports $TP = 70, TN = 20, FP = 5, FN = 5$. Compute the Accuracy.

a. 90%

b. 85%

c. 93.3%

d. 75%

36 [Intermediate] • Target: The Confusion Matrix and Core Metrics

Using the same counts ($TP = 70, FP = 5, FN = 5$), calculate the model's Precision.

a. $\frac{70}{75} \approx 93.3\%$

b. $\frac{70}{70+5} = 90\%$

c. $\frac{70}{100} = 70\%$

d. $\frac{75}{100} = 75\%$

37 [Intermediate] • Target: The Confusion Matrix and Core Metrics

Calculate the Recall (Sensitivity) for the model above.

a. $\frac{70}{75} \approx 93.3\%$

b. $\frac{70}{90} \approx 77.8\%$

c. $\frac{5}{75} \approx 6.7\%$

d. 85%

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 **[Challenging]** • **Target: The F_1-Score, Thresholding and ROC-AUC** • **Cross-Lesson: 3.01 Gradient Descent and Optimization**

Compute the F_1-score for a model having Precision $= 0.80$ and Recall $= 0.60$.

a. $\frac{2(0.8)(0.6)}{0.8+0.6} \approx 0.686$

b. 0.700

c. 0.480

d. 0.750

39 **[Challenging]** • **Target: The F_1-Score, Thresholding and ROC-AUC** • **Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

If you lower the classification probability threshold from 0.5 to 0.2, what is the typical effect on Precision and Recall?

a. Recall increases (fewer false negatives), while Precision typically decreases (more false alarms)

b. Precision increases while Recall decreases

c. Both metrics increase simultaneously

d. Neither metric changes

40 **[Challenging]** • **Target: The F_1-Score, Thresholding and ROC-AUC** • **Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

What does an Area Under the ROC Curve (ROC-AUC) of 0.50 indicate about a classifier?

a. The model has no discriminative ability (equivalent to random guessing)

b. The model is a perfect classifier

c. The model achieves 50% precision at all thresholds

d. The model is inverted

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41 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Consider a simple logistic regression model where the decision boundary is defined by the equation $-1.5x_1 + 2x_2 + 5 = 0$. For a data point (x_1, x_2), if $-1.5x_1 + 2x_2 + 5 < 0$, it is classified into Class 0. Otherwise, it is classified into Class 1. Which of the following points lies on the decision boundary?

a. $(-1, 1)$

b. $(2, -2)$

c. $(0, -2.5)$

d. $(3, 0.5)$

42 [Challenging] • Target: The Odds Ratio and Logit Formulation • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A researcher is analyzing the relationship between the number of hours a student spends on extracurricular activities per week and their final exam score in a challenging STEM course. The researcher collects data from 100 students and fits a linear regression model. The model is given by $\text{Exam Score} = 65 + 2.5 \times \text{Hours} - 0.3 \times \text{Hours}^2$, where Exam Score is the final exam score and Hours is the number of hours spent on extracurricular activities per week. The researcher believes that for a certain range of hours, there might be an optimal number of extracurricular hours that maximizes the exam score, and beyond that, the score might decrease due to time constraints. \\ \\ Consider the function representing the exam score as a function of hours: $f(h) = 65 + 2.5h - 0.3h^2$. To find the maximum exam score and the corresponding hours, calculus can be used. \\ Which of the following is the approximate number of extracurricular hours per week that would maximize the student's final exam score, assuming the model holds for $h \geq 0$?

a. 4.17 hours

b. 2.5 hours

c. 5.0 hours

d. 8.33 hours

43 [Challenging] • Target: Binary Cross-Entropy Loss • Cross-Lesson: 1.01 Vectors and Embeddings in AI

The scatterplot below displays the relationship between the number of hours (h) students spent studying for a math test and their scores (s) on that test. The equation of the least-squares regression line for this data is given by $s = 1.2h + 78.5$. If a student studied for 5 hours, what is their predicted score based on this model?

a. 91.5

b. 84.5

c. 81.5

d. 90.5

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44 [Challenging] • Target: The Confusion Matrix and Core Metrics • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider the following quadratic loss function for a regression problem: $L(y, \hat{y}) = (y - \hat{y})^2$. If the predicted value $\hat{y} = 5$ and the true value $y = 3$, what is the value of the loss function?

a. 4

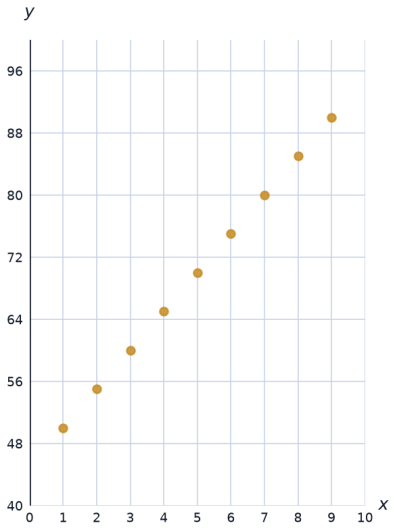
b. -2

c. 9

d. 2

45 [Challenging] • Target: The F_1-Score, Thresholding and ROC-AUC • Cross-Lesson: 3.01 Gradient Descent and Optimization

The following diagram shows a scatterplot comparing the number of hours studied per week for a math course and the final exam score for a sample of students. What is the approximate y-intercept of the line of best fit?



a. 50

b. 60

c. 70

d. 80

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